

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Winter 2025 ( Supply. )**

**Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 4**

**Subject Code & Name: BTMPE405B - SHEET METAL ENGINEERING**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Why is corrosion resistance desirable in sheet metal products?
    - a) Increases weight
    - b) Prevents rust in outdoor/wet environments like roofing
    - c) Reduces formability
    - d) Lowers conductivity
  - 2 What is a key desirable property of sheet metal materials?
    - a) High weight for stability
    - b) Good formability, corrosion resistance, and low weight
    - c) Poor weldability
    - d) Brittleness for sharp edges
  - 3 In blanking, the \_\_\_\_\_ is the useful product.
    - a) Remaining sheet after cut
    - b) Punched-out slug (blank)
    - c) Waste material only
    - d) Hole created
  - 4 In punching operation, clearance is provided on the \_\_\_\_\_ to ensure clean cuts.
    - a) Punch
    - b) Die
    - c) Sheet
    - d) Both equally
  - 5 Deep drawing is considered "deep" when:
    - a) Depth equals diameter
    - b) Depth exceeds diameter
    - c) Depth is half diameter
    - d) No flange present
  - 6  
In deep drawing, the blank holder is necessarily required
    - a) True b) False

- 7 Springback in bending occurs due to:
  - a) Plastic deformation only
  - b) Elastic recovery after load release
  - c) High temperature
  - d) Thick sheets
- 8 Progressive dies feature:
  - a) Single station for all cuts
  - b) Multiple stations; strip advances per stroke
  - c) Non-cutting operations only
  - d) None of the options
- 9 Combination dies combine:
  - a) Cutting with forming/bending
  - b) Hydraulic and mechanical presses
  - c) Only blanking/ piercing
  - d) Deep drawing only
- 10 Mechanical presses excel in:
  - a) Slow, high-force deep drawing
  - b) High-speed, repetitive shearing
  - c) Variable speed control
  - d) None of the options
- 11 In automotive body panels; the primary sheet metal process is:
  - a) Shearing only
  - b) Deep drawing and stamping
  - c) Casting
  - d) None of the options
- 12 Washers/ gaskets in appliances are typically made by:
  - a) Compound dies for blanking/piercing
  - b) Extrusion
  - c) Welding
  - d) Injection molding

Q2. Solve the following.

- A) Explain the importance of sheet metal engineering. 6
- B) Write the advantages of products made from sheet metals. 6

Q3. Solve the following.

- A) Explain the concept of blanking and punching with neat sketches. 6
- B) A circular blank of 30 mm diameter is to be cut from 2 mm thick C10 steel sheet. Determine the die and punch size. Also estimate the punch force and the Stripping Force needed. You may assume the following for the steel: Tensile strength: 410 Mpa, Shear strength: 310 Mpa. 6

Q4. Solve Any Two of the following.

- A) Explain shallow drawing of Cylindrical bodies. 6
- B) Explain the Stock Strip Layout with suitable example and explain how the material utilization can be increased with efficient stock strip layout. 6
- C) Explain the process of metal spinning. 6

- Q5. Solve Any Two of the following.
- A) Explain compound dies with neat sketch. 6
  - B) Compare the mechanical press with hydraulic press used in sheet metal stating their advantages and disadvantages. 6
  - C) Briefly explain modern developments in press tools. 6
- Q6. Solve Any Two of the following.
- A) Give any two applications of each of the following operations. 6
    - 1. Lancing 2. Perforating 3. Bending
  - B) Suggest with sheet metal operations are used to manufacture the following parts. 6
    - 1. Washbasin 2. Coin 3. Cup
  - C) Explain the sheet metal manufacturing process for any sheet metal product used in kitchens. 6

**\*\*\* End \*\*\***